PROBLEM 47

Two Repelling Particles on Concentric Wires in a Rotating Plane

Original statement · Problems.pdf, p. 32

Relative Equilibrium

We consider a system of two heavy particles of equal mass m. The first particle P is constrained to move without friction along a circular wire of radius a. The second particle Q is constrained to move without friction along a circular wire of radius $2a$. Both wires are contained in a plane Ox_1y_1 that rotates at a constant, known angular rate Ω about the axis $Oy_1 \equiv Oy_0$ with respect to a fixed reference frame S_0 with axes $Ox_0y_0z_0$ (not shown in the figure). We also define a moving reference frame S_1 with axes $Ox_1y_1z_1$. The two particles repel each other with a force that is proportional to their distance, with constant of proportionality, k.

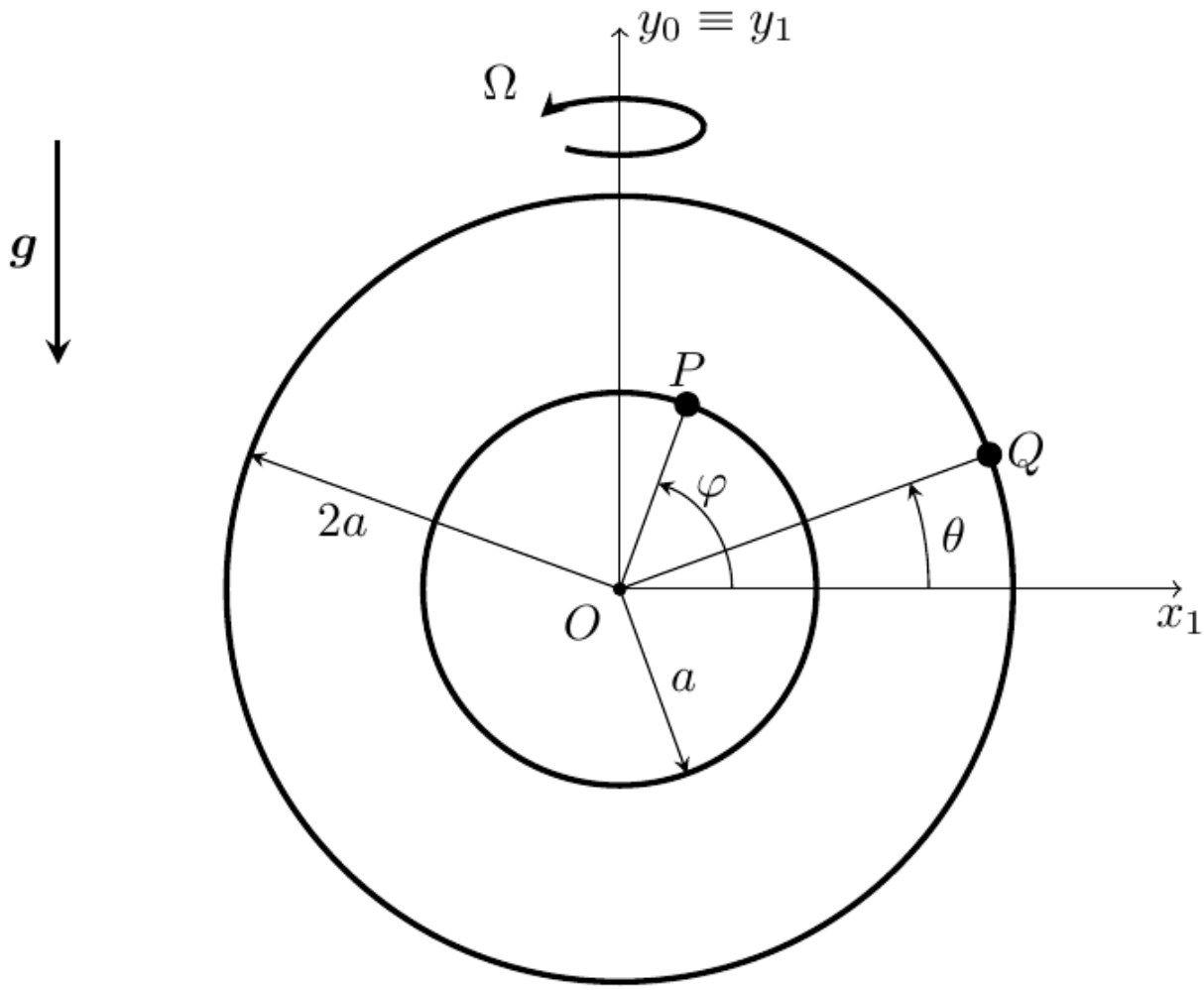


Figure: Concentric circular wires of radii a and 2a centred at O in the plane Ox1y1, particle P at angle varphi and particle Q at angle theta from x1, the plane spins with rate Omega about the vertical axis y0 = y1, gravity g downward (Problems.pdf, p. 32).

(a) How many configuration degrees of freedom has this system?

(b) How many independent components of the reaction forces on the particles need to be considered? Write down the reaction force on each particle.

(c) Write down the relative accelerations of P and Q with respect to S_1 as a function of the coordinates of the problem and their derivatives.

(d) Express the applied forces on the particles as a function of the coordinates of the problem.

(e) Express the fictitious forces of inertia on the particles as seen from S_1, as a function of the coordinates of the problem and their derivatives.

(f) Write down a system of ordinary differential equations to determine the evolution of the coordinates of the problem as a function of time, free of any other unknowns. You do not need to solve this system (since no initial conditions have been provided) but we assume that we know how to solve it, so that from this point we know the coordinates as a function of time.

(g) Determine the values of the reaction forces from the wires as a function of time.

(h) Is the mechanical energy of this system conserved in S_1? If so, write down the corresponding equation.

(i) Determine the values of $\alpha = \dfrac{ka}{mg}$ and $\beta = \dfrac{a\Omega^2}{g}$ so that $\varphi = 30°, \theta = -60°$ is an equilibrium position of the system in the relative motion.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** two heavy particles of mass m; P on a circular wire of radius a and Q on a concentric circular wire of radius $2a$, both centred at O and lying in the plane Ox_1y_1. The plane spins about the vertical axis $Oy_1 \equiv Oy_0$ with constant rate Ω. The particles repel each other with a force k times their distance.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; $\mathbf{j}_0$ is vertical upward and gravity is $-g\,\mathbf{j}_0$ (the figure draws $\mathbf{g}$ antiparallel to y_1).
  2. Smooth, bilateral wires: each reaction is perpendicular to the tangent of its wire.

3. The repulsion acts along the line PQ: on P it is $k\left(\mathbf{r}^P-\mathbf{r}^Q\right)$, on Q it is $k\left(\mathbf{r}^Q-\mathbf{r}^P\right)$ (action and reaction).
4. φ and θ are measured from Ox_1 in the positive sense about $\mathbf{k}_1$, as in the figure. The sense of Ω only affects the sign of the Coriolis terms; we take $\boldsymbol{\omega}_{10}=\Omega\,\mathbf{j}_0$ with Ω a signed constant.

- **Why S_1:** both wires are fixed in S_1, so each particle has a single angular coordinate there. The inertia forces (Notes Eq. 4.62) are the price, and with $\boldsymbol{\omega}_{10}$ constant and O fixed only the centrifugal and Coriolis forces appear.

Symbol	Meaning	SI unit
m	mass of each particle	kg
a	radius of the inner wire (outer: $2a$)	m
k	repulsion constant	N/m
Ω	constant rotation rate of the plane	rad/s
g	gravitational acceleration	m/s^2
φ,θ	angular positions of P and Q from Ox_1	rad
$\alpha=ka/(mg),\ \beta=a\Omega^2/g$	dimensionless parameters	1

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames:** $S_0=\{O;\mathcal{B}_0\}$ and $S_1=\{O;\mathcal{B}_1\}$ with common origin; S_1 is rotated by the angle Ωt about $\mathbf{j}_0=\mathbf{j}_1$ (we take $S_1\equiv S_0$ at $t=0$). A simple rotation about $\mathbf{j}_0$ gives

$$\mathbf{i}_1=\cos\Omega t\,\mathbf{i}_0-\sin\Omega t\,\mathbf{k}_0,\qquad \mathbf{j}_1=\mathbf{j}_0,\qquad \mathbf{k}_1=\sin\Omega t\,\mathbf{i}_0+\cos\Omega t\,\mathbf{k}_0$$

$$[{}_0R_1]=\begin{pmatrix}\cos\Omega t & 0 & \sin\Omega t\\ 0 & 1 & 0\\ -\sin\Omega t & 0 & \cos\Omega t\end{pmatrix}$$

- **Checks:** $\det[{}_0R_1]=\cos\Omega t\,(\cos\Omega t)+\sin\Omega t\,(\sin\Omega t)=1$ (expansion along the second row). Rows: $\cos^2\Omega t+\sin^2\Omega t=1$, 1, $\sin^2\Omega t+\cos^2\Omega t=1$; first and third rows: $-\cos\Omega t\sin\Omega t+\sin\Omega t\cos\Omega t=0$; so $[{}_0R_1][{}_0R_1]^T=I$. Also $\mathbf{i}_1\times\mathbf{j}_1=\sin\Omega t\,\mathbf{i}_0+\cos\Omega t\,\mathbf{k}_0=\mathbf{k}_1$ (right-handed).

- **Motion of S_1: $\mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$.** Check: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = -\Omega\,(\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{k}_0) = -\Omega\,\mathbf{k}_1 = \Omega\,\mathbf{j}_1 \times \mathbf{i}_1$. Since Ω and $\mathbf{j}_0$ are constant, $\boldsymbol{\alpha}_{10} = \mathbf{0}$.
- **Polar bases of the two particles** (in the plane Ox_1y_1):

$$\mathbf{e}_r^P = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1, \quad \mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1; \qquad \mathbf{e}_r^Q = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \quad \mathbf{e}_\theta = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$$

Each triad $\{\mathbf{e}_r, \mathbf{e}_\varphi$ or $\mathbf{e}_\theta, \mathbf{k}_1\}$ is orthonormal and right-handed ($\mathbf{e}_r \times \mathbf{e}_\varphi = (\cos^2\varphi + \sin^2\varphi)\,\mathbf{k}_1 = \mathbf{k}_1$). Dot products used below: $\mathbf{e}_r^Q \cdot \mathbf{e}_r^P = \cos(\varphi - \theta)$, $\mathbf{e}_r^Q \cdot \mathbf{e}_\varphi = \sin(\theta - \varphi)$, $\mathbf{e}_r^P \cdot \mathbf{e}_\theta = \sin(\varphi - \theta)$, $\mathbf{j}_1 \cdot \mathbf{e}_\varphi = \cos\varphi$, $\mathbf{j}_1 \cdot \mathbf{e}_r^P = \sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_\varphi = -\sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_r^P = \cos\varphi$ (and the same with θ for Q).

PHASE 3

Step-by-Step Mathematical Deduction

(a) Configuration degrees of freedom

> **Pedagogical Justification:** each particle on a wire (a curve) loses 2 of its 3 coordinates.

P: 3 coordinates minus 2 constraints ($z_1 = 0$, $x_1^2 + y_1^2 = a^2$) = 1. Q: 3 minus 2 ($z_1 = 0$, $x_1^2 + y_1^2 = 4a^2$) = 1. The motion of the plane is prescribed. Hence

$$\text{DOF} = 2, \qquad \text{coordinates } (\varphi, \theta)$$

(b) Reaction forces

> **Pedagogical Justification:** a smooth wire cannot push along its tangent, so its reaction lies in the normal plane of the wire, which is spanned by two independent directions; there is one unknown per removed coordinate.

Two components per particle, four in total:

$$\mathbf{N}^P = N_1\,\mathbf{e}_r^P + N_2\,\mathbf{k}_1, \qquad \mathbf{N}^Q = N_3\,\mathbf{e}_r^Q + N_4\,\mathbf{k}_1$$

(c) Relative accelerations

> **Pedagogical Justification:** $\mathbf{i}_1, \mathbf{j}_1$ are constant in S_1; by the chain rule $\left.\dfrac{d\mathbf{e}_r^P}{dt}\right|_1 = \dfrac{d\mathbf{e}_r^P}{d\varphi}\dot\varphi = \dot\varphi\,\mathbf{e}_\varphi$ and $\left.\dfrac{d\mathbf{e}_\varphi}{dt}\right|_1 = \dot\varphi\,(-\cos\varphi\,\mathbf{i}_1 - \sin\varphi\,\mathbf{j}_1) = -\dot\varphi\,\mathbf{e}_r^P$ (same for θ).

$$\mathbf{r}_1^P = a\,\mathbf{e}_r^P, \qquad \mathbf{v}_1^P = a\dot\varphi\,\mathbf{e}_\varphi, \qquad \mathbf{a}_1^P = a\ddot\varphi\,\mathbf{e}_\varphi + a\dot\varphi\left(-\dot\varphi\,\mathbf{e}_r^P\right) = a\ddot\varphi\,\mathbf{e}_\varphi - a\dot\varphi^2\,\mathbf{e}_r^P$$

$$\mathbf{r}_1^Q = 2a\,\mathbf{e}_r^Q, \qquad \mathbf{v}_1^Q = 2a\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_1^Q = 2a\ddot\theta\,\mathbf{e}_\theta - 2a\dot\theta^2\,\mathbf{e}_r^Q$$

(d) Applied forces

> **Pedagogical Justification:** the applied (actual, non-reaction) forces are the weights and the mutual repulsion; their components in each particle's polar basis are what enters the equations.

Weights: $-mg\,\mathbf{j}_1$ on each particle, i.e. $-mg\sin\varphi\,\mathbf{e}_r^P - mg\cos\varphi\,\mathbf{e}_\varphi$ on P and $-mg\sin\theta\,\mathbf{e}_r^Q - mg\cos\theta\,\mathbf{e}_\theta$ on Q.

Repulsion, with $\mathbf{r}^P - \mathbf{r}^Q = a\,\mathbf{e}_r^P - 2a\,\mathbf{e}_r^Q = a\,(\cos\varphi - 2\cos\theta)\,\mathbf{i}_1 + a\,(\sin\varphi - 2\sin\theta)\,\mathbf{j}_1$:

$$\mathbf{F}_k^P = k\left(\mathbf{r}^P - \mathbf{r}^Q\right) = ka\,[1 - 2\cos(\varphi-\theta)]\,\mathbf{e}_r^P + 2ka\sin(\varphi-\theta)\,\mathbf{e}_\varphi$$

$$\mathbf{F}_k^Q = k\left(\mathbf{r}^Q - \mathbf{r}^P\right) = ka\,[2 - \cos(\varphi-\theta)]\,\mathbf{e}_r^Q - ka\sin(\varphi-\theta)\,\mathbf{e}_\theta$$

(For P: $\mathbf{e}_r^Q\cdot\mathbf{e}_r^P = \cos(\varphi-\theta)$ and $-2a\,\mathbf{e}_r^Q\cdot\mathbf{e}_\varphi = -2a\sin(\theta-\varphi) = 2a\sin(\varphi-\theta)$. For Q: $-a\,\mathbf{e}_r^P\cdot\mathbf{e}_\theta = -a\sin(\varphi-\theta)$.)

(e) Inertia forces in S_1

> **Pedagogical Justification:** Notes Eq. 4.62 with $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$: only the centrifugal force $-m\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1)$ and the Coriolis force $-2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1$ remain (Slide 28).

Cross products with $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$. For P, $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega a\left(\cos\varphi\, \mathbf{j}_1 \times \mathbf{i}_1 + \sin\varphi\, \mathbf{j}_1 \times \mathbf{j}_1\right) = -\Omega a \cos\varphi\, \mathbf{k}_1$ and $\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = -\Omega^2 a \cos\varphi\, \mathbf{j}_1 \times \mathbf{k}_1 = -\Omega^2 a \cos\varphi\, \mathbf{i}_1$. Hence (it is $m\Omega^2 \mathbf{r}_\perp$, Slide 30):

$$\mathbf{F}_C^P = m\Omega^2 a \cos\varphi\, \mathbf{i}_1 = m\Omega^2 a \cos\varphi \left(\cos\varphi\, \mathbf{e}_r^P - \sin\varphi\, \mathbf{e}_\varphi\right), \qquad \mathbf{F}_C^Q = 2m\Omega^2 a \cos\theta \left(\cos\theta\, \mathbf{e}_r^Q - \sin\theta\, \mathbf{e}_\theta\right)$$

For the Coriolis force, $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \Omega a\dot{\varphi}\left(-\sin\varphi\, \mathbf{j}_1 \times \mathbf{i}_1 + \cos\varphi\, \mathbf{j}_1 \times \mathbf{j}_1\right) = \Omega a\dot{\varphi}\sin\varphi\, \mathbf{k}_1$, so

$$\mathbf{F}_{COR}^P = -2m\Omega a\dot{\varphi}\sin\varphi\, \mathbf{k}_1, \qquad \mathbf{F}_{COR}^Q = -4m\Omega a\dot{\theta}\sin\theta\, \mathbf{k}_1$$

The Coriolis forces are perpendicular to the plane of the wires: only the reactions N_2, N_4 can balance them.

(f) Equations of motion free of unknown reactions

> **Pedagogical Justification:** write $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) for each particle and project on its tangent ($\mathbf{e}_\varphi$, $\mathbf{e}_\theta$), which is perpendicular to both reaction components and to the Coriolis force.

P, along $\mathbf{e}_\varphi$: $ma\ddot{\varphi} = -mg\cos\varphi + 2ka\sin(\varphi - \theta) - m\Omega^2 a \sin\varphi\cos\varphi$.

Q, along $\mathbf{e}_\theta$: $2ma\ddot{\theta} = -mg\cos\theta - ka\sin(\varphi - \theta) - 2m\Omega^2 a \sin\theta\cos\theta$.

Dividing by ma and $2ma$:

$$\boxed{\ddot{\varphi} = -\frac{g}{a}\cos\varphi + \frac{2k}{m}\sin(\varphi - \theta) - \Omega^2 \sin\varphi\cos\varphi, \qquad \ddot{\theta} = -\frac{g}{2a}\cos\theta - \frac{k}{2m}\sin(\varphi - \theta) - \Omega^2 \sin\theta\cos\theta}$$

(g) Reactions as functions of time

> **Pedagogical Justification:** the remaining projections ($\mathbf{e}_r$ and $\mathbf{k}_1$) contain one reaction each; with $\varphi(t), \theta(t)$ known from (f), they give the reactions explicitly.

P, along $\mathbf{e}_r^P$: $-ma\dot{\varphi}^2 = N_1 - mg\sin\varphi + ka\left[1 - 2\cos(\varphi-\theta)\right] + m\Omega^2 a\cos^2\varphi$. Along $\mathbf{k}_1$: $0 = N_2 - 2m\Omega a\dot{\varphi}\sin\varphi$.

Q, along $\mathbf{e}_r^Q$: $-2ma\dot{\theta}^2 = N_3 - mg\sin\theta + ka\left[2 - \cos(\varphi-\theta)\right] + 2m\Omega^2 a\cos^2\theta$. Along $\mathbf{k}_1$: $0 = N_4 - 4m\Omega a\dot{\theta}\sin\theta$.

$$\boxed{\begin{aligned} N_1(t) &= -ma\dot{\varphi}^2 + mg\sin\varphi - ka\left[1 - 2\cos(\varphi-\theta)\right] - m\Omega^2 a\cos^2\varphi, \quad N_2(t) = 2m\Omega a\dot{\varphi}\sin\varphi \\ N_3(t) &= -2ma\dot{\theta}^2 + mg\sin\theta - ka\left[2 - \cos(\varphi-\theta)\right] - 2m\Omega^2 a\cos^2\theta, \quad N_4(t) = 4m\Omega a\dot{\theta}\sin\theta \end{aligned}}$$

(h) Mechanical energy in S_1

> **Pedagogical Justification:** yes, it is conserved. In S_1 the wires are fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1 = 0$; the Coriolis forces do no work (Slide 31); the Euler force and $-m\mathbf{a}_0^O$ vanish; the weights, the repulsion (a central force linear in the distance) and the centrifugal forces (constant $\boldsymbol{\omega}_{10}$, Slide 30) all derive from potentials.

Potentials in S_1: weights $mga\sin\varphi + 2mga\sin\theta$ (heights y_1); repulsion $-\frac{1}{2}k\left|\mathbf{r}^P - \mathbf{r}^Q\right|^2$ (minus sign: repulsive, so that $-\nabla_P$ gives $+k(\mathbf{r}^P - \mathbf{r}^Q)$), with

$$\left|\mathbf{r}^P - \mathbf{r}^Q\right|^2 = a^2 + 4a^2 - 4a^2\,\mathbf{e}_r^P\cdot\mathbf{e}_r^Q = 5a^2 - 4a^2\cos(\varphi-\theta)$$

and centrifugal $-\frac{1}{2}m\Omega^2\left(r_\perp^P\right)^2 - \frac{1}{2}m\Omega^2\left(r_\perp^Q\right)^2$ with $r_\perp = |x_1|$ (distance to the axis Oy_1, since $z_1 = 0$): $-\frac{1}{2}m\Omega^2 a^2\cos^2\varphi - 2m\Omega^2 a^2\cos^2\theta$. Kinetic energy: $\frac{1}{2}ma^2\dot{\varphi}^2 + \frac{1}{2}m\,4a^2\dot{\theta}^2$. Dropping the constant $-\frac{5}{2}ka^2$:

$$\boxed{E_1 = \frac{1}{2}ma^2\left(\dot{\varphi}^2 + 4\dot{\theta}^2\right) + mga\left(\sin\varphi + 2\sin\theta\right) + 2ka^2\cos(\varphi-\theta) - \frac{1}{2}m\Omega^2 a^2\left(\cos^2\varphi + 4\cos^2\theta\right) = \text{const}}$$

Check. By the chain rule ($\frac{d}{dt}\cos(\varphi-\theta) = -\sin(\varphi-\theta)(\dot{\varphi}-\dot{\theta})$, $\frac{d}{dt}\cos^2\varphi = -2\cos\varphi\sin\varphi\,\dot{\varphi}$):

$$\frac{dE_1}{dt} = \dot\varphi\left[ma^2\ddot\varphi + mga\cos\varphi - 2ka^2\sin(\varphi-\theta) + m\Omega^2a^2\sin\varphi\cos\varphi\right] + \dot\theta\left[4ma^2\ddot\theta + 2mga\cos\theta + 2ka^2\sin(\varphi-\theta) + 4m\Omega^2a^2\sin\theta\cos\theta\right]$$

and both brackets vanish by the two equations of (f) (multiplied by ma^2 and $4ma^2$), so $dE_1/dt = 0$.

(i) Parameters for the equilibrium $\varphi = 30^\circ, \theta = -60^\circ$

> **Pedagogical Justification:** a relative equilibrium has $\dot\varphi = \dot\theta = \ddot\varphi = \ddot\theta = 0$, so the right-hand sides of (f) must vanish. Dividing the first by g/a and the second by $g/(2a)$ introduces $\alpha = ka/(mg)$ and $\beta = a\Omega^2/g$.

$$0 = -\cos\varphi + 2\alpha\sin(\varphi-\theta) - \beta\sin\varphi\cos\varphi, \qquad 0 = -\cos\theta - \alpha\sin(\varphi-\theta) - 2\beta\sin\theta\cos\theta$$

With $\varphi - \theta = 90^\circ$ ($\sin = 1$), $\cos 30^\circ = \frac{\sqrt3}{2}$, $\sin 30^\circ = \frac12$, $\cos(-60^\circ) = \frac12$, $\sin(-60^\circ) = -\frac{\sqrt3}{2}$:

$$0 = -\frac{\sqrt3}{2} + 2\alpha - \frac{\sqrt3}{4}\beta, \qquad 0 = -\frac12 - \alpha + \frac{\sqrt3}{2}\beta$$

From the second, $\alpha = \frac{\sqrt3}{2}\beta - \frac12$. Substituting in the first: $-\frac{\sqrt3}{2} + \sqrt3\beta - 1 - \frac{\sqrt3}{4}\beta = 0$, i.e. $\frac{3\sqrt3}{4}\beta = 1 + \frac{\sqrt3}{2} = \frac{2+\sqrt3}{2}$, so

$$\beta = \frac{2\left(2+\sqrt3\right)}{3\sqrt3} = \frac{6+4\sqrt3}{9} \approx 1.4365, \qquad \alpha = \frac{\sqrt3}{2}\cdot\frac{6+4\sqrt3}{9} - \frac12 = \frac{12+6\sqrt3-9}{18} = \frac{1+2\sqrt3}{6} \approx 0.7440$$

$$\boxed{\alpha = \frac{ka}{mg} = \frac{1+2\sqrt3}{6} \approx 0.744, \qquad \beta = \frac{a\Omega^2}{g} = \frac{6+4\sqrt3}{9} \approx 1.4365}$$

Both values are positive, so the equilibrium is physically realisable.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Equilibrium:** at $\varphi - \theta = 90^\circ$ the radii OP and OQ are perpendicular, so the tangential part of the repulsion is as large as it can be ($2ka$ on P, ka on Q), and it balances the tangential components of the weight and of the centrifugal force. P (at 30° above Ox_1) is pushed up its wire by the repulsion from Q, while gravity and the centrifugal force pull it back; Q (at 60° below Ox_1) is pushed down by the repulsion and by gravity, and the centrifugal force (which pulls it towards $\theta = 0$, away from the axis) holds it.
- **Coriolis forces:** they are normal to the rotating plane and only load the wires (N_2, N_4); they never enter the equations of motion or the energy.
- **Limits:** $\Omega \to 0$ removes the centrifugal terms and $N_2 = N_4 = 0$; $k \to 0$ decouples the two particles into two independent rotating-ring pendulums, $\ddot{\varphi} = -\frac{g}{a}\cos\varphi - \Omega^2 \sin\varphi\cos\varphi$.
- **Dimensions:** $[g/a] = [k/m] = [\Omega^2] = \mathrm{s}^{-2} = [\ddot{\varphi}]$; $[N_1]$: $[ma\dot{\varphi}^2] = [mg] = [ka] = [m\Omega^2 a] = \mathrm{N}$; $[N_2] = [m\Omega a\dot{\varphi}] = \mathrm{N}$; $[E_1] = \mathrm{kg\,m^2/s^2} = [mga] = [ka^2] = \mathrm{J}$; $\alpha = \mathrm{N\,m^{-1}\,m/N}$ and $\beta = \mathrm{m\,s^{-2}/(m\,s^{-2})}$ are dimensionless.
- **Official key** (Problems.pdf, p. 76): $\alpha = 0.744$, $\beta = 1.4365$. Our exact values $\alpha = (1 + 2\sqrt{3})/6 = 0.74402$ and $\beta = (6 + 4\sqrt{3})/9 = 1.43647$ agree.
- **Symbolic check (sympy):** $\mathbf{a}_0$ of each particle obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1$ twice equals the Notes Eq. 3.27 decomposition; the two equations of motion, the four reactions and $dE_1/dt = 0$ are confirmed, and solving the equilibrium conditions gives exactly the values above.

Source: Problems.pdf (PDF page 32); key on PDF page 76